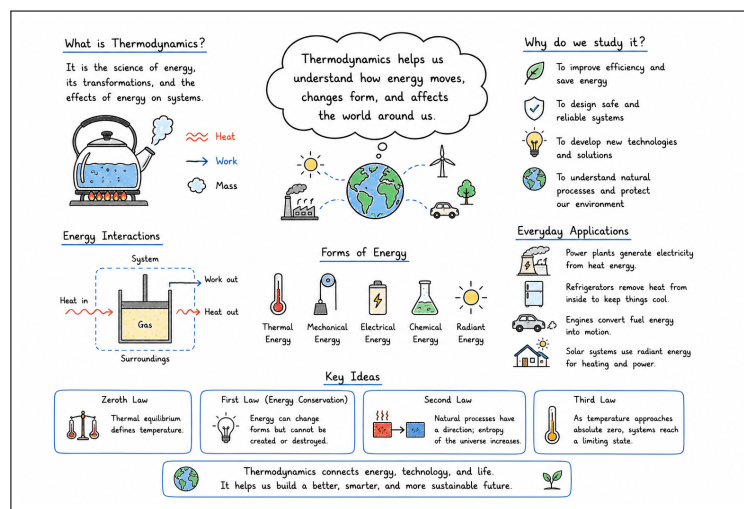




CHAPTER 1 : INTRODUCTION & BASIC CONCEPTS

This chapter introduces the fundamental concepts and vocabulary of thermodynamics needed for understanding energy and its transformations. It develops the basic ideas of systems, control volumes, properties, equilibrium, processes, and thermodynamic states. The chapter also establishes essential foundations involving dimensions, units, temperature, pressure, density, and specific properties. Together, these concepts provide the framework required for analyzing thermodynamic systems and solving engineering problems systematically.

CHAPTER TREE

- 1-1 THERMODYNAMICS AND ENERGY
Introduces thermodynamics, energy, and fundamental energy principles.
- 1-2 IMPORTANCE OF DIMENSIONS AND UNITS
Establishes dimensions, units, and commonly used measurement systems.
- 1-3 SYSTEMS AND CONTROL VOLUMES
Defines systems, surroundings, boundaries, and control volumes.
- 1-4 PROPERTIES OF A SYSTEM
Classifies system properties as intensive, extensive, and specific.
- 1-5 DENSITY AND SPECIFIC GRAVITY
Defines density, specific volume, specific gravity, and specific weight.
- 1-6 STATE AND EQUILIBRIUM
Describes thermodynamic states and different equilibrium conditions.
- 1-7 PROCESSES AND CYCLES
Defines processes, process paths, quasi-equilibrium, and cycles.
- 1-8 TEMPERATURE AND THE ZEROth LAW OF THERMODYNAMICS
Introduces temperature measurement and the zeroth law.
- 1-9 PRESSURE
Defines pressure and examines its variation within fluids.
- 1-10 PRESSURE MEASUREMENT DEVICES
Introduces instruments used for measuring atmospheric and fluid pressure.
- 1-11 PROBLEM-SOLVING TECHNIQUE
Presents a systematic approach for solving engineering problems.
- A Remark on Significant Digits
Explains appropriate precision and rounding of engineering results.

1.1 THERMODYNAMICS AND ENERGY

Thermodynamics is the science concerned with energy, energy transformations, and the interaction between energy and matter.

THERMODYNAMICS

Energy may be viewed as the **ability to cause changes**.

The word thermodynamics originates from the Greek words therme (heat) and dynamis (power), reflecting the early development of the subject around converting heat into power.

Today, thermodynamics has a much broader scope and deals with:

- Energy and its different forms.
- Energy transformations.
- Power generation.
- Refrigeration.
- Relationships among the properties of matter.

CONSERVATION OF ENERGY

The **conservation of energy principle** states that during an interaction, energy can change from one form to another, but the total amount of energy remains constant.

Energy cannot be created or destroyed; it can only change forms.

For any system, the change in its energy content is equal to the difference between the energy entering and leaving the system.

$$\Delta E = E_{\text{in}} - E_{\text{out}}$$

Example

When a rock falls from a height, its potential energy is converted into kinetic energy. The total energy is conserved even though the form of energy changes.

FIRST AND SECOND LAWS OF THERMODYNAMICS

The two fundamental laws of thermodynamics describe important characteristics of energy.

Law	Main principle
First law	Energy is conserved.
Second law	Energy has both quantity and quality.

The **first law of thermodynamics** is an expression of the conservation of energy principle and establishes energy as a thermodynamic property.

The **second law of thermodynamics** states that energy has quality as well as quantity. Actual processes occur in the direction of decreasing energy quality.

Example

A hot cup of coffee placed in a cooler environment eventually loses heat and becomes cooler. However, a cool cup of coffee in the same environment does not spontaneously become hot. Thus, energy naturally tends to be transferred from a region of higher temperature to a region of lower temperature, resulting in degradation of the usefulness or quality of energy.

MACROSCOPIC AND MICROSCOPIC APPROACHES

Matter consists of a very large number of particles called molecules. The properties observed at the macroscopic level arise from the behavior of these particles.

For example, the pressure of a gas results from momentum transfer between gas molecules and the walls of its container.

Two approaches can be used to study thermodynamics:

Approach	Basis	Characteristic
Classical thermodynamics	Macroscopic behavior	Direct and relatively simple
Statistical thermodynamics	Average microscopic behavior	More involved

Classical thermodynamics

Classical thermodynamics does not require knowledge of the behavior of individual particles. It studies measurable macroscopic quantities and provides a direct and convenient method for solving engineering problems.

Statistical thermodynamics

Statistical thermodynamics is based on the average behavior of large groups of individual particles. It is a microscopic approach and is more involved than classical thermodynamics.

In engineering applications, classical thermodynamics is generally the primary approach, while statistical thermodynamics may be used in a supporting role.

APPLICATION AREAS OF THERMODYNAMICS

Interactions between energy and matter occur throughout nature. Consequently, thermodynamics is relevant to a wide range of engineering systems and everyday phenomena.

Area	Examples
Human body	Metabolic energy conversion and heat rejection
Household systems	Heating, air conditioning, refrigerators, water heaters
Appliances	Pressure cookers, irons, computers, televisions
Transportation	Automobile engines, aircraft, rockets, jet engines
Energy systems	Conventional and nuclear power plants, solar collectors
Industrial systems	Pumps, piping networks, and other thermal systems

1.2 IMPORTANCE OF DIMENSIONS AND UNITS

Any physical quantity can be characterized by its **dimensions**. The magnitudes assigned to dimensions are called

units. Some dimensions are selected as primary or fundamental dimensions, while others are expressed in terms of the primary dimensions and are called secondary or derived dimensions.

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DIMENSIONS AND UNITS

Quantity	Type	Description
Mass, length, time, temperature	Primary dimensions	Fundamental dimensions
Velocity, energy, volume	Secondary dimensions	Expressed in terms of primary dimensions

Two unit systems are commonly used in engineering:

- **SI system:** The metric International System of Units.
- **English system:** Also known as the United States Customary System (USCS).

The SI system has a simple decimal relationship between its units and is widely used in scientific and engineering work. The English system has less systematic relationships between units. Both systems are used in engineering, although SI units are emphasized. :contentReference[oaicite:1]index=1 :contentReference[oaicite:2]index=2

SI FUNDAMENTAL DIMENSIONS AND UNITS

The SI system is based on seven fundamental quantities and their corresponding units. :contentReference[oaicite:3]index=3

Dimension	SI Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Temperature	kelvin	K
Electric current	ampere	A
Amount of light	candela	cd
Amount of matter	mole	mol

The SI system uses standard prefixes to represent multiples and submultiples of units.

Multiple	Prefix	Symbol
10^{24}	yotta	Y
10^{21}	zetta	Z
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^2	hecto	h
10^1	deka	da

Multiple	Prefix	Symbol
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a
10^{-21}	zepto	z
10^{-24}	yocto	y

The SI prefixes are standard for all units and are widely used throughout engineering. :contentReference[oaicite:4]index=4

UNIT CONVENTIONS

SI unit names are written without capitalization, even when derived from proper names. However, abbreviations of units derived from proper names are capitalized.

For example:

- The SI unit is newton, not Newton.
- Its abbreviation is N.
- A unit abbreviation is not pluralized.
- No period is used after a unit abbreviation unless it occurs at the end of a sentence.

Thus, 5 m or 5 meters is correct, while 5 ms or 5 meter is incorrect. :contentReference[oaicite:5]index=5

Dimension	SI Unit	English Unit
Length	1 m	3.28084 ft
Mass	1 kg	2.20462 lbm
Time	1 s	1 s
Force	1 N	0.224809 lbf
Pressure	1 kPa	0.145038 psi
Energy	1 kJ	0.947817 Btu
Power	1 kW	1.34102 hp
Volume	1 m^3	35.3147 ft^3
Specific Volume	$1 \text{ m}^3/\text{kg}$	$16.0185 \text{ ft}^3/\text{lbm}$
Density	$1 \text{ kg}/\text{m}^3$	$0.062428 \text{ lbm}/\text{ft}^3$
Specific Energy	$1 \text{ kJ}/\text{kg}$	$0.429923 \text{ Btu}/\text{lbm}$
Specific Weight	$1 \text{ kN}/\text{m}^3$	$6.36588 \text{ lbf}/\text{ft}^3$

DIMENSIONAL HOMOGENEITY

An equation used in engineering must be **dimensionally homogeneous**. This means every term in an equation must have the same units.

If quantities with different units are added or subtracted, an error has occurred.

Dimensional analysis can therefore be used to:

- Check the validity of equations.
- Detect errors in calculations.
- Verify that terms being added or subtracted have compatible dimensions.

A formula that is not dimensionally homogeneous is definitely incorrect. However, dimensional homogeneity alone does not guarantee that a formula is physically correct.

Example

If an equation contains terms representing energy, every term added to or subtracted from those terms must also have units of energy.

UNITY CONVERSION RATIOS

Nonprimary dimensions can be formed from combinations of primary dimensions. Similarly, non-primary units can be formed from combinations of primary units.

A **unity conversion ratio** is a ratio of two equivalent quantities and is exactly equal to one.

For example:

$$\frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} = 1$$

and

$$\frac{1 \text{ lbf}}{32.174 \text{ lbm} \cdot \text{ft/s}^2} = 1$$

Since these ratios are equal to one, they can be inserted into calculations without changing the physical value of a quantity. They are used to convert units and maintain dimensional consistency.

Important points:

- A unity conversion ratio is exactly equal to 1.
- Its inverse is also exactly equal to 1.
- The ratio should be arranged so that unwanted units cancel.
- Unity conversion ratios can be used to convert between different unit systems.

The use of unity conversion ratios is preferred over introducing the gravitational constant g_c merely to force units to match.

1.3 SYSTEMS AND CONTROL VOLUMES

SYSTEM, SURROUNDINGS, AND BOUNDARY

A **system** is defined as a quantity of matter or a region in space chosen for study.

The **surroundings** are the mass or region outside the system.

The **boundary** is the real or imaginary surface that separates the system from its surroundings.

Term	Definition
System	Quantity of matter or region in space chosen for study
Surroundings	Mass or region outside the system
Boundary	Surface separating the system from its surroundings

The boundary may be **fixed** or **movable**. It is the contact surface shared by the system and its surroundings.

Mathematically, the boundary has zero thickness. Therefore, it can neither contain mass nor occupy any volume.

TYPES OF SYSTEMS

Systems are classified as **closed systems** or **open systems** depending on whether a fixed mass or a fixed region in space is selected for study.

System	Mass Transfer	Energy Transfer
Closed system	Not allowed	Allowed
Isolated system	Not allowed	Not allowed
Open system	Allowed	Allowed

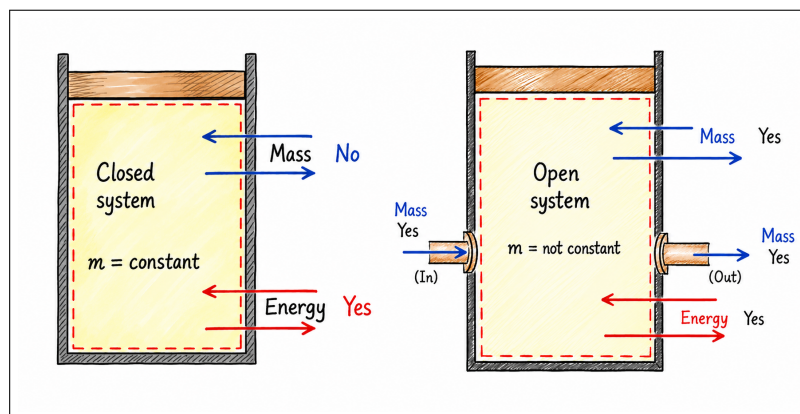


Figure 1: Types of System

CLOSED SYSTEM

A **closed system**, also called a **control mass**, consists of a fixed amount of mass.

No mass can cross the boundary of a closed system, but energy in the form of heat or work can cross the boundary.

The volume of a closed system does not necessarily have to remain fixed.

Isolated System

An **isolated system** is a special type of closed system in which neither mass nor energy is allowed to cross the boundary.

$$m = \text{constant}$$

EXAMPLE OF A CLOSED SYSTEM

Consider a piston–cylinder device containing gas.

If the gas is selected as the system, the inner surfaces of the piston and cylinder form the boundary.

Since no mass crosses this boundary, the gas forms a closed system.

Energy may cross the boundary, for example, when the gas is heated. The boundary may also move because the piston can move.

Part	Description
System	Gas enclosed in the cylinder
Boundary	Inner surfaces of piston and cylinder
Surroundings	Everything outside the gas
Mass transfer	None
Energy transfer	Possible
Boundary	May move with the piston

OPEN SYSTEM

An **open system**, also called a **control volume**, is a properly selected region in space.

A control volume usually encloses a device through which mass flows, such as:

- Compressor
- Turbine
- Nozzle

Both **mass** and **energy** can cross the boundary of a control volume.

A large number of engineering problems involve mass flow into and out of a system. Such systems are therefore more conveniently modeled as control volumes rather than control masses.

SELECTION OF A CONTROL VOLUME

In general, any arbitrary region in space can be selected as a control volume.

There are no strict rules for selecting a control volume. However, a proper choice can greatly simplify the analysis.

For example, when analyzing the flow of air through a nozzle, the region within the nozzle is a convenient choice for the control volume.

Control Surface

The boundary of a control volume is called a **control surface**.

A control surface may be:

- Real.
- Imaginary.
- Fixed.
- Moving.

For a nozzle, the inner wall of the nozzle forms the real part of the control surface, while the entrance and exit areas form imaginary parts of the control surface.

FIXED AND MOVING CONTROL VOLUMES

A control volume may be fixed in size and shape or may involve a moving boundary.

Type	Description
Fixed control volume	Size and shape remain constant
Moving control volume	Boundary can move
Real boundary	Corresponds to a physical surface
Imaginary boundary	Used where no physical surface exists

Most control volumes have fixed boundaries and therefore do not involve moving boundaries.

A control volume can involve heat and work interactions in addition to mass interaction.

EXAMPLE OF AN OPEN SYSTEM

Consider a water heater in which cold water enters the tank and hot water leaves the tank.

Since hot water continuously leaves the tank and is replaced by cold water, it is not convenient to choose a fixed mass as the system.

Instead, the region enclosed by the interior surfaces of the tank can be selected as the control volume.

The cold water entering the tank and the hot water leaving the tank represent mass crossing the control surface.

Part	Description
Control volume	Interior region of the water heater
Control surface	Interior surfaces of the tank
Mass entering	Cold water
Mass leaving	Hot water
Energy interaction	Heat transferred to the water

The water heater therefore provides an example of an open system with mass entering and leaving the control volume.

IMPORTANCE OF PROPER SYSTEM SELECTION

The system under study must be defined carefully in an engineering analysis.

In simple problems, the system may be obvious and its definition may appear unnecessary. In more complicated problems, however, selecting an appropriate system can greatly simplify the analysis.

Closed System vs Control Volume

Feature	Closed System	Control Volume
Amount of matter	Fixed	May change
Mass crossing boundary	No	Yes
Energy crossing boundary	Yes	Yes
Boundary	Real or imaginary	Real or imaginary
Boundary motion	May be fixed or movable	May be fixed or movable
Typical example	Piston-cylinder	Turbine or nozzle

1.4 PROPERTIES OF A SYSTEM

PROPERTIES OF A SYSTEM

Any characteristic of a system is called a **property**. Common thermodynamic properties include pressure P , temperature T , volume V , and mass m . Other properties include viscosity, thermal conductivity, modulus of elasticity, thermal expansion coefficient, electrical resistivity, velocity, and elevation.

Properties are classified as **intensive** or **extensive** depending on whether they depend on the amount of matter present in the system.

Intensive Properties

Intensive properties are independent of the mass or size of the system.

Examples:

Property	Examples
Intensive	Temperature, pressure, density

Extensive Properties

Extensive properties depend on the size or amount of matter in the system.

Examples:

Property	Examples
Extensive	Mass, volume, total momentum

Identifying Intensive and Extensive Properties

An easy criterion is to imagine dividing the system into two equal parts:

Property type	Effect of division
Intensive	Value remains unchanged
Extensive	Value becomes proportional to the amount of matter

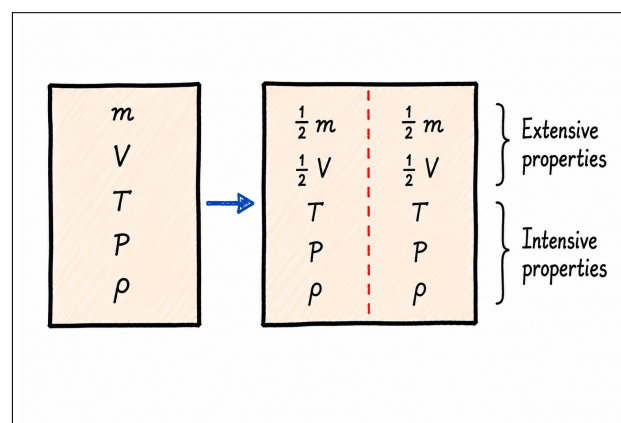


Figure 2: Test to identify type of property

SPECIFIC PROPERTIES

An extensive property expressed per unit mass is called a **specific property**.

Specific properties are intensive because dividing an extensive property by mass removes its dependence on the size of the system.

For specific volume,

$$v = \frac{V}{m}$$

where:

- v = specific volume
- V = total volume
- m = mass

Specific total energy is another example:

$$e = \frac{E}{m}$$

where E is the total energy of the system.

CONTINUUM

Matter is actually composed of discrete atoms and molecules. In thermodynamic analysis, however, it is usually convenient to ignore the molecular structure and regard matter as a continuous, homogeneous substance called a **continuum**.

The continuum idealization allows:

- properties to be treated as point functions,
- properties to vary continuously throughout the system, and
- abrupt molecular-scale variations to be neglected.

The continuum assumption is valid when the characteristic dimension of the system is much larger than the distance between molecules or, more specifically, the mean free path of the molecules.

For oxygen at 1 atm and 20°C:

$$d_{O_2} \approx 3 \times 10^{-10} \text{ m}$$

$$\lambda \approx 6.3 \times 10^{-8} \text{ m}$$

where λ is the mean free path. Even in a volume of 1 mm³, there are approximately

$$3 \times 10^{16}$$

molecules of oxygen under these conditions.

Thus, for ordinary engineering systems, the molecular structure can usually be ignored and the substance can be treated as a continuum.

At very high vacuum or at very high elevations, the mean free path becomes large relative to the characteristic dimension of the system. In such cases, the continuum approximation may fail and rarefied-gas-flow theory must be used.

1.5 DENSITY & SPECIFIC GRAVITY

DENSITY

Density is defined as the mass per unit volume of a substance.

$$\rho = \frac{m}{V}$$

where ρ is density, m is mass, and V is volume.

The SI unit of density is kg/m^3 .

For a differential volume element, density can also be expressed as

$$\rho = \frac{\delta m}{\delta V}$$

The density of a substance generally depends on temperature and pressure. For most gases, density is approximately proportional to pressure and inversely proportional to temperature.

Liquids and solids are essentially incompressible, so their density changes only slightly with pressure. Their density is generally more sensitive to temperature than to pressure.

SPECIFIC VOLUME

Specific volume is defined as the volume occupied per unit mass. It is the reciprocal of density.

$$v = \frac{V}{m} = \frac{1}{\rho}$$

The SI unit of specific volume is m^3/kg .

Thus,

$$\rho v = 1$$

Density–Specific Volume Relationship

Quantity	Definition	SI Unit
Density, ρ	$\frac{m}{V}$	kg/m^3
Specific volume, v	$\frac{V}{m}$	m^3/kg

SPECIFIC GRAVITY

Specific gravity, also called relative density, is the ratio of the density of a substance to the density of a standard substance at a specified temperature.

For most applications in this section, water at 4°C is used as the standard, with

$$\rho_{\text{H}_2\text{O}} = 1000 \text{ kg/m}^3$$

The specific gravity is therefore

$$SG = \frac{\rho}{\rho_{H_2O}}$$

Specific gravity is **dimensionless** because it is a ratio of two densities having the same units.

Interpretation of Specific Gravity

Specific Gravity	Interpretation
$SG < 1$	Substance is less dense than water
$SG = 1$	Density is equal to that of water
$SG > 1$	Substance is denser than water

For example, mercury has a specific gravity of 13.6 at 0°C. Hence,

$$\rho_{Hg} = 13.6(1000) = 13600 \text{ kg/m}^3$$

In SI units, the numerical value of specific gravity is also equal to the numerical value of density when density is expressed in g/cm^3 or kg/L .

Since

$$1 \text{ g/cm}^3 = 1 \text{ kg/L} = 1000 \text{ kg/m}^3$$

a substance with $SG = 13.6$ has a density of 13.6 g/cm^3 or 13.6 kg/L .

SPECIFIC WEIGHT

Specific weight is defined as the weight of a substance per unit volume.

$$\gamma_s = \rho g$$

where γ_s is the specific weight, ρ is the density, and g is the gravitational acceleration.

The SI unit of specific weight is N/m^3 .

1.6 STATE AND EQUILIBRIUM

STATE OF A SYSTEM

The **state** of a system is the condition of the system described by the values of its properties. When a system is not undergoing any change, its properties can be measured or calculated throughout the system, providing a complete description of its state.

At a given state, all the properties of the system have fixed values. If the value of even one property changes, the system changes to a different state.

Equilibrium

Thermodynamics deals primarily with **equilibrium states**. Equilibrium implies a state of balance in which there are no unbalanced potentials or driving forces within the system.

A system in equilibrium does not undergo any change when it is isolated from its surroundings.

A system is in **thermodynamic equilibrium** only when all the relevant types of equilibrium are satisfied.

TYPES OF EQUILIBRIUM

Type	Condition
Thermal equilibrium	No temperature difference within the system
Mechanical equilibrium	No pressure changes with time at any point
Phase equilibrium	Mass of each phase remains at an equilibrium level
Chemical equilibrium	Chemical composition does not change with time

Condition for Thermodynamic Equilibrium

A system cannot be considered to be in thermodynamic equilibrium unless all the relevant equilibrium requirements are simultaneously satisfied.

THE STATE POSTULATE

Although the state of a system is described by its properties, it is not necessary to specify all the properties to completely fix the state. Once a sufficient number of independent properties are specified, the remaining properties automatically assume definite values.

State Postulate

The state of a simple compressible system is completely specified by **two independent, intensive properties**.

Simple Compressible System

A simple compressible system is a system in which electrical, magnetic, gravitational, motion, and surface-tension effects are absent or negligible.

These effects are generally negligible in most engineering problems. If any such effect is significant, an additional property must be specified for each important effect.

For example, if gravitational effects are significant, the elevation z must be specified in addition to the two properties required by the state postulate.

Independent Properties

The two properties used to specify the state must be **independent**.

Two properties are independent if one property can be varied while the other is held constant.

Temperature and specific volume are always independent properties. Therefore, for a simple compressible system,

$$T \text{ and } v$$

can be used together to completely specify the state.

Temperature and pressure behave differently depending on the phase condition. For a single-phase system, T and P are independent properties.

For a multiphase system, however, temperature and pressure can become dependent properties. During a phase-change process,

$$T = f(P)$$

Thus, temperature and pressure alone are not sufficient to fix the state of a two-phase system.

Example of State Specification

For nitrogen, two independent intensive properties such as temperature and specific volume can completely specify the state:

$$T = 25^\circ\text{C}$$

$$v = 0.9 \text{ m}^3/\text{kg}$$

Hence, no additional independent intensive property is required for a simple compressible nitrogen system under the assumptions of the state postulate.

1.7 PROCESSES AND CYCLES

PROCESS

Any change that a system undergoes from one equilibrium state to another is called a **process**. The series of equilibrium states through which the system passes during a process is called the **path of the process**.

A process is completely described by specifying:

- the initial state,
- the final state,
- the path followed by the system, and
- the interactions between the system and its surroundings.

Quasi-Equilibrium Process

A process during which the system remains infinitesimally close to an equilibrium state at all times is called a **quasi-equilibrium** or **quasi-static process**.

It can be viewed as a sufficiently slow process that allows the system to adjust internally so that the properties remain nearly uniform throughout the system.

Example :

Consider a gas in a piston-cylinder device.

During a **very rapid compression**, the molecules near the piston do not have sufficient time to redistribute. A high-pressure region develops near the piston, and the system is not in equilibrium.

During a **slow compression**, the molecules have enough time to redistribute. The pressure remains nearly uniform throughout the cylinder, so the process can be treated as quasi-equilibrium.

Importance of Quasi-Equilibrium Processes

A quasi-equilibrium process is an idealization rather than an exact representation of a real process. However, many actual processes can be approximated by quasi-equilibrium processes with negligible error.

Engineers are interested in quasi-equilibrium processes because:

- they are easier to analyze, and
- work-producing devices deliver the most work when operating through quasi-equilibrium processes.

Therefore, quasi-equilibrium processes provide useful standards against which actual processes can be compared.

PROCESS DIAGRAMS AND PROCESS PATH

Thermodynamic processes can be represented graphically by using thermodynamic properties as coordinates.

Common properties used as coordinates are:

Coordinate	Property
T	Temperature
P	Pressure
V or v	Volume or specific volume

For example, a compression process can be represented on a P-V diagram, with the process path connecting the initial and final states.

The process path has physical significance for a **quasi-equilibrium process** because every point on the path represents an equilibrium state.

For a **nonquasi-equilibrium process**, the entire system cannot be described by a single equilibrium state at each instant. Therefore, a process path for the system as a whole cannot be defined. Such a process is represented by a dashed line between the initial and final states.

COMMON TYPES OF PROCESSES

The prefix **iso-** indicates that a particular property remains constant during a process.

Isothermal Process

An isothermal process is a process during which the temperature remains constant.

$$T = \text{constant}$$

Isobaric Process

An isobaric process is a process during which the pressure remains constant.

$$P = \text{constant}$$

Isochoric Process

An isochoric process, also called an isometric process, is a process during which the specific volume remains constant.

$$v = \text{constant}$$

CYCLE

A system is said to undergo a **cycle** if it returns to its initial state at the end of the process.

Thus, for a cycle,

$$\text{Initial state} = \text{Final state}$$

Although the system may undergo several processes while completing a cycle, the final state is identical to the initial state.

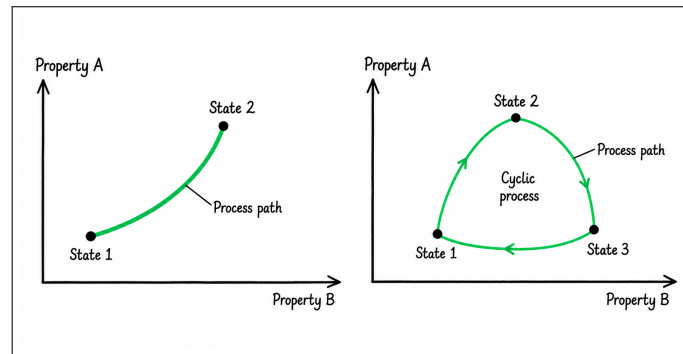


Figure 3: Process & Cycle

STEADY-FLOW PROCESS

Term	Meaning
Steady	No change with time
Uniform	No change with location

A large number of engineering devices operate for long periods under essentially constant conditions and can be modeled as **steady-flow devices**.

A **steady-flow process** is a process in which fluid flows through a control volume steadily.

During a steady-flow process:

- fluid properties may vary from point to point within the control volume,
- at any fixed point, the fluid properties do not change with time,
- the volume of the control volume remains constant,
- the mass contained in the control volume remains constant, and
- the total energy content of the control volume remains constant.

Thus,

$$V = \text{constant}$$

$$m = \text{constant}$$

$$E = \text{constant}$$

Steady-flow conditions can be closely approximated in devices intended for continuous operation, such as turbines, pumps, and boilers.

1.8 TEMPERATURE AND THE ZEROth LAW OF THERMODYNAMICS

TEMPERATURE

Temperature is a measure of the thermal state of a substance and is commonly associated with the degree of hotness or coldness. Human senses provide only a qualitative indication of temperature and cannot be relied upon for accurate measurement.

Accurate temperature measurement is possible because certain properties of materials change with temperature in a repeatable and predictable manner. For example, a mercury-in-glass thermometer operates on the expansion of mercury with temperature.

Thermal Equilibrium

When two bodies at different temperatures are brought into contact, heat is transferred from the body at higher temperature to the body at lower temperature.

This transfer continues until both bodies reach the same temperature. At that point, heat transfer stops and the bodies are said to be in **thermal equilibrium**.

The only requirement for thermal equilibrium is equality of temperature.

ZEROth LAW OF THERMODYNAMICS

The **zeroth law of thermodynamics** states:

Zeroth Law

If two bodies are each in thermal equilibrium with a third body, then they are in thermal equilibrium with each other.

The zeroth law provides the basis for temperature measurement. A thermometer can be regarded as the third body. If two bodies produce the same temperature reading on the thermometer, they are in thermal equilibrium with each other even when they are not in direct contact.

Temperature Measurement

The zeroth law establishes temperature as a property that can be compared between systems.

Thus, a thermometer can be used to determine whether two bodies are at the same temperature without requiring them to be in direct contact.

TEMPERATURE SCALES

Temperature scales provide a common basis for assigning numerical values to temperature.

They are based on reproducible reference states. Two historically important reference states are:

Reference state	Description
Ice point	Ice and water in equilibrium at 1 atm
Steam point	Liquid water and water vapor in equilibrium at 1 atm

Temperature Scales Comparison

Scale	Ice Point	Steam Point
Celsius	0°C	100°C
Fahrenheit	32°F	212°F

THERMODYNAMIC TEMPERATURE SCALES

A thermodynamic temperature scale should be independent of the properties of any particular substance.

The thermodynamic temperature scale in the SI system is the **Kelvin scale**, while the corresponding scale in the English system is the **Rankine scale**.

Kelvin Scale

The thermodynamic temperature unit in the SI system is the **kelvin**, symbol K.

The degree symbol is not used with kelvin. Thus, the correct notation is 300 K, not 300°K.

The lowest possible temperature is **absolute zero**:

$$T = 0 \text{ K}$$

Absolute zero corresponds to

$$T = -273.15^\circ\text{C}$$

Rankine Scale

The thermodynamic temperature scale in the English system is the **Rankine scale**.

Its temperature unit is the **rankine**, symbol R.

IDEAL-GAS TEMPERATURE SCALE

An ideal-gas temperature scale is nearly identical to the Kelvin scale.

It can be measured using a constant-volume gas thermometer, which consists of a rigid vessel containing a gas at low pressure.

At sufficiently low pressure and constant volume, the temperature of the gas is proportional to its pressure.

The relationship can be expressed as

$$T = a + bP$$

where a and b are experimentally determined constants.

The constants are determined by measuring the gas pressure at two reproducible reference temperatures.

If the ice point is assigned 0°C and the steam point is assigned 100°C, the resulting gas-temperature scale agrees with the Celsius scale.

On extrapolation to zero pressure, the temperature is

$$T = -273.15^\circ\text{C}$$

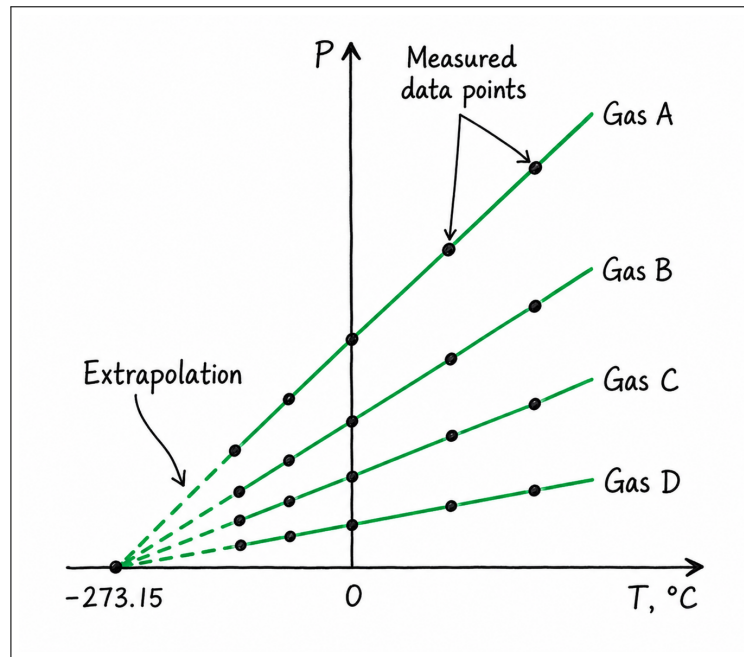
This corresponds to absolute zero.

By assigning zero temperature at zero absolute pressure, the relation becomes

$$T = bP$$

This provides an absolute gas-temperature scale.

The absolute gas-temperature scale agrees with the thermodynamic temperature scale over the range in which the gas thermometer can be used, although it cannot be used at extremely low or extremely high temperatures because of phenomena such as condensation, dissociation, and ionization.



TEMPERATURE CONVERSION RELATIONS

The Kelvin and Celsius scales are related by

$$T(K) = T(^{\circ}C) + 273.15$$

The Rankine and Fahrenheit scales are related by

$$T(R) = T(^{\circ}F) + 459.67$$

The Rankine and Kelvin scales are related by

$$T(R) = 1.8T(K)$$

The Fahrenheit and Celsius scales are related by

$$T(^{\circ}F) = 1.8T(^{\circ}C) + 32$$

For ordinary engineering calculations, the constants 273.15 and 459.67 are sometimes rounded to 273 and 460, respectively.

TEMPERATURE DIFFERENCES

The magnitude of one kelvin is identical to the magnitude of one degree Celsius.

Therefore,

$$\Delta T(\text{K}) = \Delta T(^{\circ}\text{C})$$

Similarly, the magnitude of one rankine is identical to the magnitude of one degree Fahrenheit.

Therefore,

$$\Delta T(\text{R}) = \Delta T(^{\circ}\text{F})$$

The relationship between the magnitudes of the units is

$$1 \text{ K} = 1^{\circ}\text{C} = 1.8 \text{ R} = 1.8^{\circ}\text{F}$$

Thus, increasing a temperature by 10°C is equivalent to increasing it by 10 K.

Temperature in Thermodynamic Relations

When a thermodynamic relation contains a **temperature difference**, either K or $^{\circ}\text{C}$ may be used because their interval magnitudes are identical.

However, when a relation contains the **temperature itself**, the absolute temperature scale must be used. In the SI system, this means using K rather than $^{\circ}\text{C}$.

When in doubt, Kelvin is the safe choice for thermodynamic equations involving absolute temperature.

1.9 PRESSURE

DEFINITION OF PRESSURE

Pressure is defined as the **normal force exerted by a fluid per unit area**.

$$P = \frac{F_n}{A}$$

where F_n is the normal force acting on the surface and A is the surface area.

Pressure is a **scalar quantity**. In fluids, pressure acts normal to a surface.

The SI unit of pressure is the pascal:

$$1 \text{ Pa} = 1 \text{ N/m}^2$$

The pascal is small for most engineering applications, so larger pressure units are commonly used.

Common Pressure Units

Unit	Equivalent
1 kPa	10^3 Pa
1 MPa	10^6 Pa
1 bar	10^5 Pa
1 atm	101325 Pa
1 kgf/cm ²	$9.807 \times 10^4 \text{ Pa}$
1 psi	1 lbf/in ²

Other useful conversions are

$$1 \text{ bar} = 100 \text{ kPa} = 0.1 \text{ MPa}$$

$$1 \text{ atm} = 101.325 \text{ kPa} = 1.01325 \text{ bar}$$

$$1 \text{ atm} = 14.696 \text{ psi}$$

$$1 \text{ kgf/cm}^2 = 0.9807 \text{ bar} = 0.9679 \text{ atm} = 14.223 \text{ psi}$$

Pressure on Solid Surfaces

Pressure can also be used for the normal stress exerted on a solid surface.

For example, if a person's weight is 150 lbf and the total area of the feet is 50 in²,

$$P = \frac{150}{50} = 3 \text{ psi}$$

Reducing the contact area increases the pressure for the same force.

This explains why:

- snowshoes prevent a person from sinking deeply into snow, and
- a sharp knife cuts more easily than a blunt knife.

ABSOLUTE, GAGE, AND VACUUM PRESSURE

The pressure at a given location measured relative to absolute vacuum is called **absolute pressure**. Absolute vacuum corresponds to zero absolute pressure.

Most pressure-measuring devices are calibrated to read zero at the local atmospheric pressure. Therefore, they indicate the difference between the absolute pressure and the atmospheric pressure.

This difference is called **gage pressure**.

Absolute Pressure

Absolute pressure is measured relative to absolute vacuum.

$$P_{abs} = 0$$

at absolute vacuum.

Gage Pressure

Gage pressure is the pressure measured relative to the local atmospheric pressure.

$$P_{gage} = P_{abs} - P_{atm}$$

Gage pressure may be positive or negative.

Vacuum Pressure

When the absolute pressure is below atmospheric pressure, the difference is called **vacuum pressure**.

$$P_{vac} = P_{atm} - P_{abs}$$

Pressure Relationships

$$P_{abs} = P_{atm} + P_{gage}$$

$$P_{abs} = P_{atm} - P_{vac}$$

Pressure	Reference	Typical Meaning
Absolute	Absolute vacuum	Actual pressure
Gage	Atmospheric pressure	Pressure above or below atmosphere
Vacuum	Atmospheric pressure	Amount by which pressure is below atmosphere

In thermodynamic relations and tables, pressure is generally taken to mean **absolute pressure** unless otherwise specified.

The suffixes "a" and "g" may be attached to pressure units:

Notation	Meaning
psia	Pounds per square inch absolute
psig	Pounds per square inch gage

Example :

A vacuum gage connected to a chamber reads 5.8 psi at a location where the atmospheric pressure is 14.5 psi.

The absolute pressure is

$$P_{abs} = P_{atm} - P_{vac} = 14.5 - 5.8 = 8.7 \text{ psi}$$

Thus, the chamber pressure is 8.7 psia.

VARIATION OF PRESSURE WITH DEPTH

In a fluid at rest, pressure does not change in the horizontal direction.

Therefore, all points lying on the same horizontal plane within the same continuous fluid have the same pressure.

Pressure does, however, vary in the vertical direction because of gravity.

Pressure Increase with Depth

Pressure increases with depth because a deeper fluid layer supports a greater amount of fluid above it.

For a constant-density fluid, the pressure difference between two points is

$$\Delta P = P_2 - P_1 = -\rho g \Delta z$$

where $\Delta z = z_2 - z_1$.

Since pressure increases as elevation decreases, a convenient form is

$$P_{\text{below}} = P_{\text{above}} + \rho g |\Delta z|$$

Using specific weight,

$$P_{\text{below}} = P_{\text{above}} + \gamma_s |\Delta z|$$

where

$$\gamma_s = \rho g$$

Pressure Head

For a given fluid, the vertical distance corresponding to a pressure difference is called the **pressure head**.

Pressure therefore depends on:

- the vertical distance between the points,
- the density of the fluid, and
- the gravitational acceleration.

Pressure Variation in Gases

For gases, the density is usually very low. Therefore, over small to moderate elevation changes, the variation of pressure with height is negligible.

Consequently, the pressure inside a gas-filled tank or within a room can often be approximated as uniform.

For fluids whose density varies significantly with elevation,

$$\frac{dP}{dz} = -\rho g$$

The pressure difference between two points is then

$$\Delta P = P_2 - P_1 = - \int_1^2 \rho g \, dz$$

For constant density and constant gravitational acceleration, this expression reduces to the hydrostatic relation.

HYDROSTATIC PRESSURE AND CONTAINER SHAPE

For a fluid at rest, the pressure is independent of the shape or cross-sectional geometry of the container.

It changes only with vertical position.

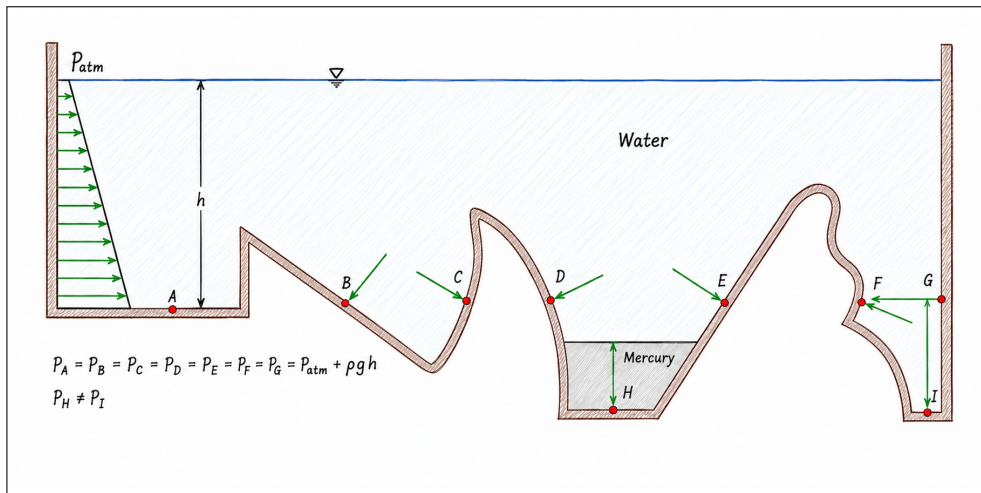
Therefore, all points on the same horizontal plane in the same interconnected fluid have the same pressure.

$$P_A = P_B = P_C = P_D = P_E = P_F = P_G$$

provided that the points are at the same elevation and are interconnected through the same continuous static fluid.

However, points at the same depth do not necessarily have the same pressure when they belong to different fluids or are not interconnected through the same fluid.

The pressure force exerted by a fluid is always **normal to the surface** on which it acts.



PASCAL'S LAW

A consequence of pressure being uniform in the horizontal direction is that when pressure is applied to a confined fluid, the pressure increases by the same amount throughout the fluid.

This is known as **Pascal's law**.

Pascal's law forms the operating principle of hydraulic devices such as hydraulic lifts, hydraulic jacks, and hydraulic brakes.

For two connected pistons at the same level,

$$P_1 = P_2$$

Since

$$P_1 = \frac{F_1}{A_1}$$

and

$$P_2 = \frac{F_2}{A_2}$$

we obtain

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

Therefore,

$$\frac{F_2}{F_1} = \frac{A_2}{A_1}$$

The ratio

$$\frac{A_2}{A_1}$$

is called the **ideal mechanical advantage** of a hydraulic lift.

Thus, a larger piston can produce a larger output force for a relatively small input force.

Example :

For a hydraulic jack with

$$\frac{A_2}{A_1} = 100$$

the ideal mechanical advantage is 100.

Therefore,

$$F_2 = 100F_1$$

A person can therefore lift a much larger load by applying a relatively small force to the smaller piston.

1.10 PRESSURE MEASUREMENT DEVICES

THE BAROMETER

A **barometer** is a device used to measure atmospheric pressure. Therefore, atmospheric pressure is also called **barometric pressure**.

The basic barometer consists of a mercury-filled tube inverted into a container of mercury that is open to the atmosphere. The space above the mercury column contains only a very small amount of mercury vapor and can be treated as a vacuum.

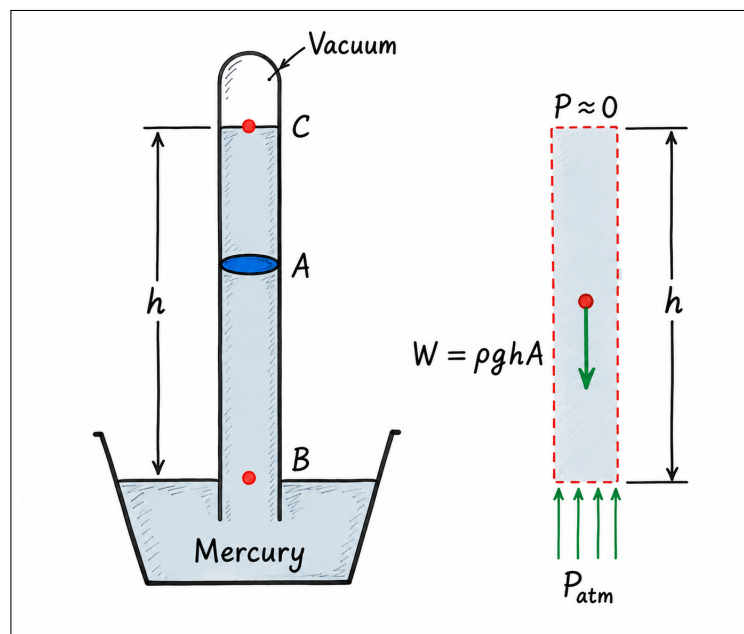
At equilibrium, the atmospheric pressure supports the mercury column. A force balance gives

$$P_{\text{atm}} = \rho gh$$

where:

- P_{atm} = atmospheric pressure,
- ρ = density of mercury,
- g = local gravitational acceleration,
- h = height of the mercury column.

The height of the mercury column is independent of the length and cross-sectional area of the tube, provided the tube is sufficiently large to make capillary effects negligible.



Standard Atmospheric Pressure

A standard atmosphere is defined as the pressure produced by a mercury column 760 mm high at 0°C under standard gravitational acceleration.

$$1 \text{ atm} = 760 \text{ mmHg}$$

Since millimeters of mercury are also called torr,

$$1 \text{ atm} = 760 \text{ torr}$$

and

$$1 \text{ torr} \approx 133.3 \text{ Pa}$$

If water were used instead of mercury, a water column of approximately 10.3 m would be required to produce standard atmospheric pressure.

Variation of Atmospheric Pressure with Elevation

Atmospheric pressure decreases as elevation increases because the weight of the air above a location decreases.

The atmospheric pressure is approximately:

Elevation	Atmospheric Pressure
Sea level	101.325 kPa
1000 m	89.88 kPa
2000 m	79.50 kPa
5000 m	54.05 kPa
10000 m	26.5 kPa
20000 m	5.53 kPa

Atmospheric pressure at a location is determined by the weight of the air above that location per unit area. It therefore changes with both elevation and weather conditions.

The decrease in atmospheric pressure with elevation has practical effects:

- water boils at a lower temperature at high elevations,
- less oxygen is available per unit volume of air,
- internal-combustion engines produce less power unless compensated, and
- aircraft experience changes in lift and drag because of the lower air density.

Example :

A barometer reads 740 mmHg at a location where

$$g = 9.805 \text{ m/s}^2$$

and the density of mercury at the specified temperature is

$$\rho_{\text{Hg}} = 13570 \text{ kg/m}^3$$

The atmospheric pressure is

$$P_{\text{atm}} = \rho gh$$

$$P_{\text{atm}} = (13570)(9.805)(0.740) = 98.5 \text{ kPa}$$

Thus, the barometer reading can be converted directly into atmospheric pressure using the hydrostatic relation.

THE MANOMETER

A **manometer** is a pressure-measuring device that uses a liquid column to determine a pressure difference.

Manometers are commonly used to measure **small and moderate pressure differences**.

A simple manometer consists of a glass or plastic U-tube containing one or more fluids such as:

- mercury,
- water,
- alcohol, or
- oil.

Heavy fluids such as mercury are useful when large pressure differences must be measured because a smaller column height can then produce the required pressure difference.

The basic principle follows from hydrostatics:

$$\Delta P = \rho gh$$

Pressure increases when moving downward through a fluid and decreases when moving upward. For a manometer open to the atmosphere,

$$P_2 = P_{\text{atm}} + \rho gh$$

where h is the difference in fluid-column elevation.

The cross-sectional area of the tube does not affect the pressure difference produced by a given height difference. However, the tube diameter should be sufficiently large to minimize capillary effects.

Important Rules for Manometer Analysis

When moving through a manometer:

Movement	Pressure change
Downward through a fluid	Add ρgh
Upward through a fluid	Subtract ρgh
Horizontally in the same continuous fluid	No pressure change

Thus, the pressure at an unknown point can be found by starting from a point of known pressure and adding or subtracting the appropriate ρgh terms.

Inclined Manometer

An **inclined manometer** uses a slanted tube to increase the displacement of the liquid column for a given pressure difference.

This increases the **resolution** and makes small pressure differences easier to measure accurately.

MULTIFLUID MANOMETERS

Some manometers contain several immiscible fluids with different densities.

The analysis is based on three rules:

1. The pressure change across a fluid column of height h is ρgh .
2. Pressure increases downward and decreases upward.
3. Two points at the same elevation in the same continuous fluid at rest have the same pressure.

For stacked fluid layers, starting from a point of known pressure and moving downward through each layer gives

$$P_{\text{bottom}} = P_{\text{top}} + \rho_1 gh_1 + \rho_2 gh_2 + \rho_3 gh_3$$

If all fluids have the same density, this reduces to

$$P_{\text{bottom}} = P_{\text{top}} + \rho g(h_1 + h_2 + h_3)$$

Differential Manometer

A differential manometer measures the pressure difference between two specified points, often across a flow device such as a valve, heat exchanger, or other flow resistance.

For the arrangement shown in the section, the pressure difference is

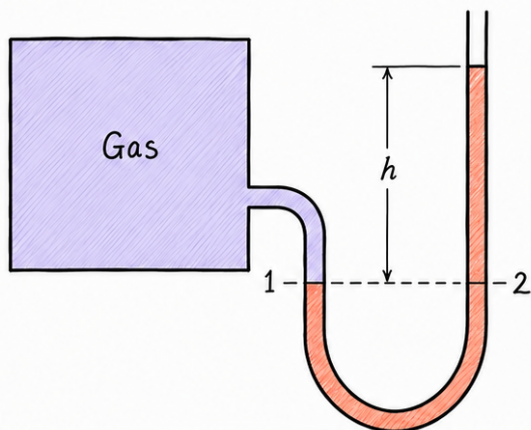
$$P_1 - P_2 = (\rho_2 - \rho_1)gh$$

where:

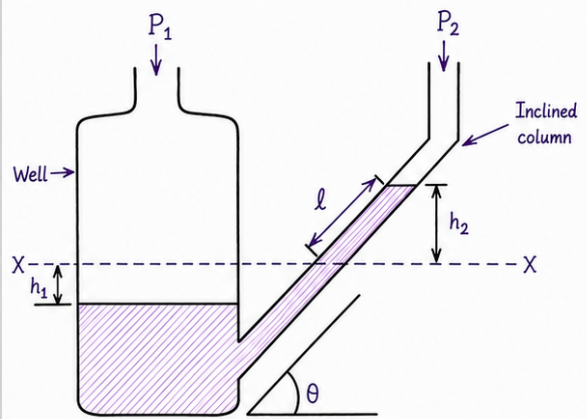
- ρ_1 = density of the flowing fluid,
- ρ_2 = density of the manometer fluid,
- h = differential fluid height.

For a gas flowing in the pipe, ρ_1 is much smaller than ρ_2 , so

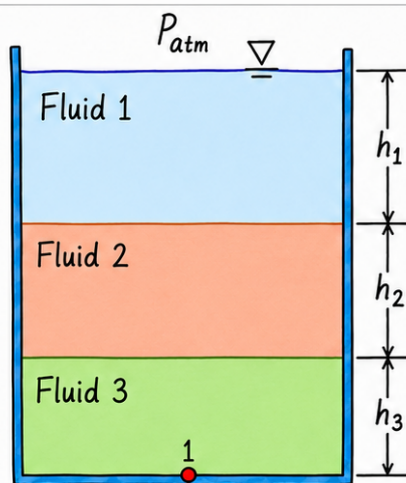
$$P_1 - P_2 \approx \rho_2 gh$$



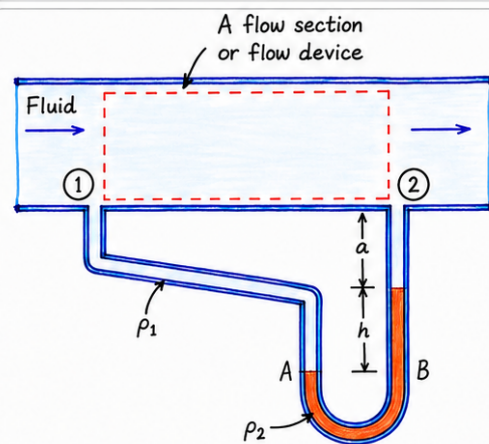
Basic Manometer



Inclined Manometer



Multifluid Manometer



Differential Manometer

1.11 PROBLEM-SOLVING TECHNIQUE

SYSTEMATIC PROBLEM SOLVING

Solving engineering problems, especially complicated ones, requires a **systematic approach**. A step-by-step method reduces a complicated problem to a series of simpler tasks and helps avoid common errors.

The recommended problem-solving procedure consists of seven steps.

Step 1: Problem Statement

State the problem briefly in your own words.

The problem statement should include:

- the key information given,
- the quantities that must be determined, and
- the main objective of the analysis.

This ensures that the physical problem and its objectives are understood before beginning the solution.

Step 2: Schematic

Draw a realistic sketch of the physical system and include the relevant information directly on the figure.

The schematic should show the important features of the system and may include:

- dimensions,
- known properties,
- unknown quantities,
- energy interactions,
- mass interactions, and
- properties that remain constant during the process.

The sketch does not need to be elaborate, but it should represent the physical system clearly.

Step 3: Assumptions and Approximations

State the assumptions and approximations used to simplify the problem.

Questionable assumptions should be justified, and reasonable values may be assumed for missing information when necessary.

For example, if no atmospheric pressure is specified, 1 atm may be assumed in some situations. However, the effect of elevation should be considered when appropriate because atmospheric pressure decreases with elevation.

Step 4: Physical Laws

Apply all relevant physical laws and principles to the system.

Examples include:

- conservation of mass,
- conservation of energy, and
- other fundamental physical relations.

The region or system to which each physical law is applied must first be identified clearly.

For example, the increase in velocity of water flowing through a nozzle can be analyzed by applying conservation of mass between the inlet and outlet.

Step 5: Properties

Determine the unknown properties at known states that are required for the solution.

These properties may be obtained from:

- property relations, or
- thermodynamic property tables.

Properties should be listed separately, and their source should be identified when appropriate.

Step 6: Calculations

Substitute the known quantities into the simplified physical relations and calculate the required unknowns.

Special attention should be given to:

- units,
- unit cancellations, and
- appropriate significant digits.

A dimensional quantity without a unit is meaningless. Intermediate calculations should retain sufficient digits, while the final result should be reported with an appropriate number of significant digits.

Step 7: Reasoning, Verification, and Discussion

The final results should be checked to determine whether they are physically reasonable and intuitive.

The analysis should include:

- verification of questionable assumptions,
- repetition of calculations that produce unreasonable results,

- interpretation of the significance of the results,
- conclusions and recommendations where appropriate, and
- limitations under which the results are valid.

The results should not be applied to situations in which the underlying assumptions are invalid.

IMPORTANCE OF ORGANIZATION

Engineering solutions are also a form of communication. Therefore, **neatness, organization, completeness, and visual clarity** are important.

A well-organized solution helps reveal errors and inconsistencies. Skipping steps merely to save time can lead to additional errors and wasted effort.

Not every problem requires every step to be written explicitly. Some steps may be unnecessary for simple problems. Nevertheless, a logical and orderly approach is strongly recommended.

Many difficulties in solving engineering problems arise not from lack of knowledge, but from a lack of organization.

SIGNIFICANT DIGITS

Engineering calculations are meaningful only to the precision of the measured data. Reporting more digits than justified gives a **false impression of accuracy**. Therefore, the final answer should never contain more significant digits than the least precise quantity used in the calculation.

DEFINITION

Significant digits are the digits in a measured quantity that carry meaning regarding its precision. For example,

$$3.75 \text{ L}$$

contains **three significant digits** because the digits 3, 7, and 5 are all known to be meaningful.

WHY SIGNIFICANT DIGITS MATTER

Suppose a container has

$$V = 3.75 \text{ L}, \quad \rho = 0.845 \text{ kg/L}.$$

A calculator gives

$$m = V\rho = 3.75 \times 0.845 = 3.16875 \text{ kg}.$$

Although mathematically correct, writing

$$3.16875 \text{ kg}$$

is misleading because both the volume and density are known only to **three significant digits**. Hence the result cannot be more accurate than three significant digits.

Therefore,

$m = 3.17 \text{ kg}$

is the correct engineering answer.

WHY CAN'T WE KEEP MORE DIGITS

The value

$$3.75 \text{ L}$$

does not mean the volume is exactly 3.750000 L.

Instead, it means the measurement is accurate only to the hundredths place.

Possible actual values include

$$3.746, 3.750, 3.753, \dots$$

because all round to

3.75.

Similarly,

0.845 kg/L

does not imply infinite precision.

Thus multiplying these measurements cannot produce a result accurate to six significant digits.

ENGINEERING PRACTICE

When performing calculations,

- keep all calculator digits during intermediate steps,
- round only the final answer,
- report only the number of significant digits justified by the given data.

MEASUREMENT UNCERTAINTY

Every experimentally measured quantity contains some uncertainty.

For example,

- if density has an uncertainty of 2%,
- then any mass calculated using that density also has approximately 2% uncertainty.

Thus extremely precise numerical answers are generally meaningless because the measurements themselves are not exact.

ENGINEERING APPROXIMATIONS

Engineers often intentionally introduce small approximations to simplify calculations.

Example:

Instead of using the exact density of water at 75°C,

$$\rho = 975 \text{ kg/m}^3,$$

it is common to use

$$\rho = 1000 \text{ kg/m}^3.$$

This introduces only about a 2.5% error, which is acceptable in many engineering calculations because other experimental uncertainties are usually of comparable magnitude. Therefore, rounding final answers to a reasonable number of significant digits is both acceptable and recommended.

PRACTICAL RULES

1. Never report more significant digits than justified by the least precise input.
2. Retain full calculator precision during intermediate calculations.
3. Round only the final answer.
4. More digits do **not** imply greater accuracy.
5. Experimental uncertainty limits the precision of every engineering calculation.

Question:

In the context of significant digits, if I say 3.75, can the actual value range from 3.746 to 3.755, basically $\pm$ half of a thousandth?

Answer:

Not exactly. Since 3.75 is rounded to two decimal places (or three significant figures), the last retained digit is in the hundredths place, so the maximum rounding error is half of one hundredth:

$$\pm \frac{0.01}{2} = \pm 0.005.$$

Thus, the actual value lies within

$$3.75 \pm 0.005.$$

This corresponds to the interval

$$3.745 \leq x < 3.755.$$

The lower bound is inclusive because 3.745 rounds to 3.75, while the upper bound is exclusive because 3.755 rounds to 3.76.

Therefore, the correct range is

$$\boxed{3.745 \leq x < 3.755.}$$

The general rule is that the uncertainty associated with a rounded number is equal to **half the value of the least significant digit**.